

Subar MAHDI

Mechanical with Electrical Engineering Student

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📁 My Portfolio

PROFILE

Mechanical and Electrical Engineering student at Cardiff University with international robotics internship experience at USP Center for Robotics — CRob. Practical experience in CAD, physical prototyping, sensor integration and experimental testing, alongside a contribution to a published engineering scoping review.

ENGINEERING EXPERIENCE

Robotics Engineer Intern — USP Center for Robotics — CRob Jun 2026 – Sep 2026

São Carlos, Brazil

- › Developed a simplified manual two-jaw gripper using Fusion 360 CAD and 3D printing, progressing through V1 and V2 physical prototypes
- › Explored component geometry, assembly and jaw movement, connecting CAD models with physical mechanism behaviour
- › Participated in hands-on robot interaction and laboratory demonstrations involving robotic platforms including Spot
- › Gained introductory exposure to ROS 2 workflows and NVIDIA Isaac within the lab environment, learning how commands, robot-state feedback and simulation relate to physical systems
- › Developed practical understanding of robotics research workflows alongside researchers, with the broader CRob research collaboration continuing after the internship

Mechanical and Electrical Team Member — Bristol Racing (Formula Student) Sep 2025 – Jul 2026

- › Contributed to mechanical and electrical development of a scaled RC vehicle within Bristol Racing
- › Supported drivetrain packaging, steering concepts, CAD-based layouts and wiring arrangements
- › Considered manufacturing and assembly constraints for laser-cut and 3D-printed components

Junior Analyst — Bristol Academic Research Society Sep 2025 – Jul 2026

- › Contributed to the published review *Multifunctional Self-Healing Structural Composites for Load Bearing Satellite Applications : A Qualitative Scoping Review*
- › Reviewed technical literature on self-healing composites and helped communicate trade-offs in mechanical performance and suitability for satellite structures
- › Credited as a Junior Analyst; read the published paper

PROJECTS

Smart Thermal Temperature Monitoring System

Built and tested an Arduino-based monitoring system integrating a DS18B20 temperature sensor, AMG8833 thermal array, LCD display and LED alerts. Investigated the system's response to a 30 °C threshold, ambient stability and distance sensitivity. Compared point sensing with spatial thermal imaging and produced a technical dissertation and research poster.

Smart Lamp — Arduino-Based Electromechanical System

Built and tested an Arduino-based smart lamp incorporating motion and sound sensors, LED output and switching components. Integrated the hardware and investigated sensor-triggered responses, wiring, power delivery and component connections through iterative testing.

PROFESSIONAL EXPERIENCE

Programmatic Advertising Lead — Global Media Organisations May 2021 – Oct 2025

- › Led delivery and optimisation of programmatic advertising campaigns across global media platforms
- › Analysed performance data, investigated delivery issues and improved operational workflows
- › Coordinated technical and commercial stakeholders to meet demanding delivery deadlines

TECHNICAL SKILLS

- › **CAD** : Fusion 360 — Beginner; Autodesk Inventor — Beginner
- › **Robotics** : ROS 2 Workflows — Introductory Exposure
- › **Electronics** : Arduino IDE / Arduino project exposure; sensor integration; hardware testing
- › **Prototyping** : 3D printing; mechanical assembly
- › **Analysis & Research** : Excel; experimental testing and data acquisition; technical reporting; literature review

EDUCATION

Cardiff University

2026 – 2030

BEng Mechanical and Electrical Engineering with a Year in Industry (H211)

University of Bristol

2025 – 2026

Foundation Year in Science, Engineering and Mathematics — CertHE, completed 2026